Status	Finished
Started	Friday, 16 October 2026, 10:09 PM
Completed	Friday, 16 October 2026, 10:13 PM
Duration	3 mins 45 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is small and unsorted ✓
- b. When the list is large and sorted
- c. When the list is large and unsorted
- d. When the list is small and sorted

Question **2**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

- a. Interpolation search
- b. Hash search
- c. Binary search
- d. Linear search ✓

Question **3**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

- a. Linear
- b. Both (1) & (3)
- c. Hash
- d. Binary ✓

Question **4**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 99

b. 89 and 94

c. 90 and 94

d. 90 and 99 ✓

Question **5**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Finding

b. Discovering

c. Mining

d. Searching ✓

Question **6**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The list is sorted

b. The search continues in the left or right sublist ✓

c. The search stops

d. The search continues from the beginning

Question **7**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The last element is returned

b. An error is raised

c. The search is considered unsuccessful ✓

d. The first element is returned

Question **8**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O👎 ✓
- b. O(log n)
- c. O(n log n)
- d. O(1)

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

- a. The list must be sorted
- b. There must be a mechanism to delete and/or insert elements in the list ✓
- c. Number values should only be present

  Number values should only be present
- d. There should be direct access to the middle element in any sublist

Question **10**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

- a. O(1) ✓
- b. O👎
- c. O(n log n)
- d. O(log n)

Question **11**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Both (1) & (3)
- b. Binary search
- c. Hash search
- d. Linear search ✓

Question **12**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(n log n)
b. O(log n) ✓
c. O🥐
d. O(1)

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Dynamic search
b. Hashing
c. Linear search ✓
d. Binary search

Question **14**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 99 ✓
b. 89 and 94
c. 94 and 99
d. 90 and 100

Question **15**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is the last element in the array
b. Item is the last element in the array or item is not there at all
c. When the item is not the array at all
d. When the item is somewhere in the middle of the array ✓